

A Flexible Trading Strategy for Different Investor Attributes in Mutual Funds

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Abstract

We now live in the age where the real interest rate is negative. How to keep the value of assets through the financial investment becomes an import and urgent issue for each modern man. Whereas the mutual fund has been one of the popular investment vehicles, we develop strategies for investing in funds. In addition to seek the high return on investment (ROI), we attempt to reduce the risk simultaneously, which is the target a well-developed investment strategy should achieve. Experimental results exhibit that our improvement on the buy-and-sell strategy can accomplish the high ROI, and also reduce the amount of transactions significantly, which saves transaction fees accordingly.

Moreover, an investor can input parameters according to the investing attribute of himself, where these parameters guide the actions of our buy-and-sell strategy to fulfil the risk attitudes of investors. Extensive simulation results show that our strategy for mutual fund investment can not only make a satisfactory profit, but also meet investor's attribute. For investors who want to invest in funds but are hard to select proper ones, our work could give satisfactory references for investing decisions.
keyword: *mutual fund, investor attribute, investment strategy, genetic algorithm*

1 Introduction

We now live in the age where the real interest rate is negative. After a period of time, the

real purchasing power of our money would reduce if we just deposit them in the bank. How to keep the value of assets through the financial investment becomes an import and urgent issue for each modern man. A mutual fund is an investment vehicle made up of a pool of assets gathered from the multitude for investing in securities such as stocks, bonds, futures and money markets. Fund investors entrust professional managers with manipulating funds, and jointly assume the investment risk, profit and loss. Over the last two decades, mutual fund has been one of popular investment vehicles due to several excellences, such as professional manipulation, risk diversification and small investment. According to ICI statistics [1], the U.S. mutual fund market remained the largest in 2011, accounting for 49 percent of the \$23.8 trillion in mutual fund assets worldwide and 90.4 million individuals owning funds.

ICI also reported that the number of funds in the U.S. mutual fund industry grew up to 8,684 at year-end 2011 [1]. Though fund sponsors merge or liquidate funds that are no longer attractive to investors, they also create new funds constantly to meet investors' demands. Among the various funds, investors select potential ones in light of several factors, e.g., market situations, investment objects, fund indicators, assets and attributes of themselves. Despite a number of auxiliary indicators, rules and tools, making a portfolio still presents a challenge to many investors, and recent studies proceed to overcome the challenge [2, 3, 4]. In this paper, we propose an investment strategy based on genetic algorithms to obtain a high profit and reduce the risk concurrently. Moreover, our strategy is flexible because it can be guided to show the investor's attribute through

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user-specified parameters.

In addition, mutual fund investors have to pay additional fees to financial service firms for the front-end and back-end load, management costs, and other value-added services. Previous works address the effect of fund fees on performance and its relevance to financial planners [5]. A high transaction cost could depress the investment performance, while no doubt increases the revenue of selling agents. Khorana et al. examined the mutual fund fees around the world in 2002, and presented a worldwide average management fee of 0.9% and 2.5% front-end loads for domestic equity funds [5]. Accordingly, our work also considers the transaction fee and sets to 3% for convenient comparison to the work of [3]. In this paper, we make the idea of [3] more clear, and then, propose a flexible investment strategy to improve their approach and consider investors' attributes based on genetic algorithms.

The rest of this paper is organized as follows. Section 2 introduces related works. Subsequently, our improvements and investment strategy are presented in Section 3. Section 4 makes a series of experimental studies, and finally, Section 5 draws our conclusion.

2 Related Works

Genetic algorithms (GAs) play an important role in studying complex adaptive systems and optimization problems based on employing a combination of search, adaptation, and optimization techniques to mimic the process of natural selection [6]. A *chromosome* in GA represents a solution candidate, and is encoded as a character string from a given alphabet set. The set of chromosomes processed simultaneously by GA constitutes a *population*. Traditionally, GA starts with a population of randomly generated chromosomes, and then delivers them to the evaluation of a *fitness function* defined to measure the solution quality for the specific problem. New chromosomes are created by selecting promising candidates and recombining them through various evolutionary operators, such as *selection*, *crossover* and *mutation*. Figure 1 shows the principle of a GA.

There have been a lot of works devoted to improve investment returns by using GAs. The first one could be [7], where a GA was applied to learn optimal technical trading rules with S&P 500 daily prices from 1928 to 1995. [8, 9] used GA to design technical trading systems, and pro-

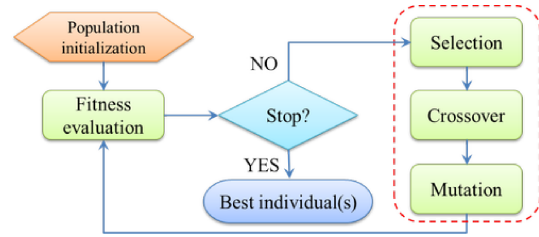


Figure 1: Principle of a genetic algorithm.

vided evidence to support the forecasting capacity of certain technical indicators in the Madrid Stock Market. Kaucic proposed a multi-objective GA to be a learning method for combining a set of signals into a unique trading signal [10], while Cheng et al. presented hybrid forecasting model of rough sets theory and GA to predict stock price trends [11]. Applying GA to develop investment strategies continues to flourish, and more recently, a number of works tried to improve the investment returns by concentrating on different parameters and technical analysis using GA [12, 13, 14, 15]. On the other hand, [16] gave a positive answer to whether GA can find profitable investment strategies based on prior stock returns and earnings information over two market states. By considering the over-learning issue of evolutionary computations, Yeh et al. proposed four important findings on the level of decision-making model and system through their empirical study [17].

Tsai et al. designed several indicators and a GA mechanism to invest in equity funds with A-type and U.S. dollar [3]. By a 10-year simulation from 1999 to 2008, they reported that an annual profit higher than 10% can be achieved. There are three main steps in their investment strategy (Figure 2): First, they defined the so-called *global trend indicator* to measure the trend of all considered funds. Second, the *monitoring indicator* was derived to evaluate the situation of financial market, i.e., *bull* or *bear*, and trigger buying or selling signals accordingly. Third, when receiving signals, GA was invoked to select funds according to their past profitability. In this paper, we follow the three steps to make our investment strategy but improve the buy and sell strategies in the bull and bear markets, respectively.

3 Fund Investment Strategy

Tsai et al. proposed an investment strategy to achieve a high annual return when considering the

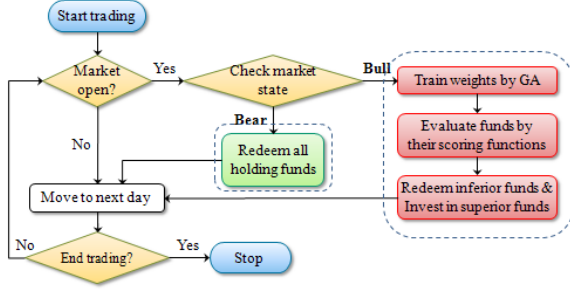


Figure 2: Investment strategy flowchart [3].

transaction fee [3]. In spite of a satisfactory profit, their investment strategy needs to be enhanced toward the investing behavior of a rational player and maximal expected revenue. The fatal drawback in their work is to redeem all fund as soon as the sell signal is triggered (see Figure 2), which could result in heavy trades and excessive transaction cost. In addition, their global trend indicator has no way to discriminate between small and dramatic movements up. Here we try to overcome the problems.

The evolution of *moving averages* and the successful applications of GAs in constructing trading systems suggest that one can develop good investment strategies by observing the rate of overall return over a given period [18]. The arithmetic average of the last days over a time period T is called *simple moving average* (SMA), which is the most basic of technical indicators. With time series data, SMA helps to smooth over short-term fluctuations and highlight long-term trends or cycles. A common strategy is to issue buying (*resp.* selling) signals when the short-term SMAs cross up (*resp.* down) the long-term ones by at least a pre-specified percentage band. SMA formulation is given as follows, where $price(t)$ represents the price of day t and T is a given time period:

$$SMA_T(t) = \frac{1}{T} \sum_{i=0}^{T-1} price(t-i),$$

In what follows, our investment strategy is introduced step by step. First, we use a modified indicator to show the global trend of market, and then, section 3.2 presents when to trigger buy-and-sell signals. After receiving signals, section 3.3 exhibits how to select profitable funds according to the historical performances of funds. Finally, we show the buying and selling strategies in sections 3.4 and 3.5, respectively, to reduce transaction costs, achieve a high profit, and also meet

different attributes of investors.

3.1 Global Trend Indicator

The *global trend indicator* (GTI) designed by [3] is used to determine the market trend. We follow this indicator but revise its formula. Let F be the set of considered funds. The *net asset value* (NAV) of a fund x at day t is represented by $NAV(x, t)$. To evaluate the performance of a fund x over a time period T , investors often use the *return of investment* (ROI) of x , denoted as $ROI_T(x, t)$, from the day t backing to a period T . If T is one day, its formula is given by

$$ROI_{1D}(x, t) = \frac{NAV(x, t)}{NAV(x, t-1)} - 1.$$

For computing $GTI(t)$ at day t , Tsai et al. first divided F into two disjoint groups of rising and falling funds, next normalized the numbers of the two groups, and lastly multiplied two normalized terms by $GTI(t-1)$ [3]. However, their GTI calculation has two fatal drawbacks, which could result in a failed program execution or an erroneous judgment of financial trend. The first is that $GTI(t)$ can be zero always. This is caused by $GTI(t) = 0$ once $ROI_{1D}(x, t) \leq 0$ for all $x \in F$, i.e., the NAVs of all considered funds are dropped, and $GTI(t+1)$ comes from the product of $GTI(t)$ and a number. The other defect is that Tsai et al. treat all increases of a fund's NAV as the same thing, i.e., they have no way to discriminate between small and dramatic movements up. Therefore, their strategy takes the same action for the boosts of 1 and 10 to the funds' NAVs. To attack the two drawbacks, we use the *NAV ratio* $NR_T(x, t)$ instead of $ROI_T(x, t)$, and propose our GTI formula as follows:

$$NR_{1D}(x, t) = \frac{NAV(x, t)}{NAV(x, t-1)},$$

$$GTI(t) = GTI(t-1) \times \frac{\sum_{x \in F} NR_{1D}(x, t)}{|F|},$$

where $|F|$ is the number of considered funds. Our GTI shall not equal to zero unless the NAVs of all funds are dropped to zero at the same time, and it is also a sensitive indicator for reflecting both sharp increases and decreases. Moreover, the base value of $GTI(0)$ is set to 100, identical to the setting of [3].

3.2 Generate Buy-and-Sell Signals

The idea of [3] to generate buy-and-sell signals is mainly based on the moving-average crossover by the *monitoring indicator* (MI), which is derived from SMAs on their GTI values. Here we adopt the MI design but use our GTI definition instead. There are five moving-average indicators, $SMA_\delta(t)$ for $\delta \in \{1, 3, 5, 7, 9\}$, involved in the following $MI_T(t)$ formula for a specific time unit T .

$$\begin{aligned} MI_T^{win}(t) &= \frac{1}{20} \sum_{1 \leq i < j \leq 5} \{j - i | SMA_{2i-1}(t) > SMA_{2j-1}(t)\} \\ MI_T^{lose}(t) &= -\frac{1}{20} \sum_{1 \leq i < j \leq 5} \{j - i | SMA_{2i-1}(t) \leq SMA_{2j-1}(t)\} \\ MI_T(t) &= MI_T^{win}(t) + MI_T^{lose}(t) \end{aligned}$$

In the above formulas, $MI_T^{win}(t)$ and $MI_T^{lose}(t)$ can be regarded as scoring functions, i.e., the former wins 1 point if $SMA_1(t) > SMA_3(t)$; otherwise, the latter loses 1 point. The rest may be deduced by analogy. The reason why we divide the sum by 20 is for the normalization. Note that the time unit T represents the period of sampling, and its value could be $1W$, $2W$, $1M$ and so on. For example, $SMA_3(t)$ in $MI_{1W}(t)$ is the average of $GTI(t)$, $GTI(t-7)$ and $GTI(t-14)$. Subsequently, Tsai et al. used MI to derive buy-and-sell signals. For the buying signal, they sampled the values of monthly GTI, whereas for two selling signals, the values of weekly GTI were sampled. Let $signal(t)$ be the buy-and-sell signal at day t , and it can then be defined as the equation (1).

At day t , $signal(t)$ triggers the buying or selling signal if the corresponding condition is met; otherwise, it returns nothing. In a bull market, a selling signal means that a bear market would be coming, and subsequently invokes the selling strategy introduced in section 3.5. On the contrary, $signal(t) = buy$ in a bear market implies the coming of a bull market. At this moment, we use the predefined buying strategy introduced later to find superior funds for the profitable investment.

3.3 Fund Selection

We pick out N good funds from F as our investment objects when a buying signal appears, where N is a user-specified parameter for the maximal size of himself portfolio list. A common way to

estimate the potential profitability of a fund is the historical achievement of its type on the market. To meet different demands of investors, we further divide the historical data of a fund into *long-term* and *short-term* parts, and our GA approach is applied to the two parts individually.

$$signal(t) = \begin{cases} buy & \text{if } MI_{1M}(t) = 1, \\ sell & \text{if } ((MI_{1W}(t-14) = 1) \text{ or } (MI_{1W}(t-7) = 1)) \\ & \text{and } ((SMA_1(t) < SMA_9(t)) \text{ or } (SMA_1(t-7) < SMA_9(t-7))), \\ sell & \text{if } ((MI_{2W}(t-28) = 1) \text{ or } (MI_{2W}(t-14) = 1)) \\ & \text{and } ((SMA_1(t) < SMA_9(t)) \text{ or } (SMA_1(t-14) < SMA_9(t-14))), \\ nothing & \text{otherwise.} \end{cases} \quad (1)$$

At first, the notation $rank(NR_T(x, t))$ represents the rank of a fund x at day t backing to a time period T among all considered funds. Taking the performances over different time periods into account, the long-term score function $Lscore(x, t)$ is then defined as follows:

$$Lscore(x, t) = \sum_{T_L} \frac{W_{T_L}(t)}{rank(NR_{T_L}(x, t))},$$

where $T_L \in \{3M, 6M, 9M, 1Y, 2Y, 3Y, 4Y, 5Y\}$ and $W_{T_L}(t) \in [0, 1]$ is the weight of x 's performance during T_L . The score function can be further regarded as the fitness function of GA. In the straightforward decision for appropriate weights, GA is invoked with each chromosome consisting of eight weights where each one is encoded by six bits. The *roulette wheel selection method* [19], a widespread sampling technique in GAs, is employed to select individuals probabilistically based on their scores. Through $Lscore(x, t)$, one can easily rank considered funds and pick up top ones as his investment objects.

On the other hand, we have the short-term consideration of funds' achievements for liquid investments. In general, the mutual fund is regarded as the long-term investment vehicle, and hence, investors are expected to hold funds for a while. Lots of mutual fund corporations even punish investors for the frequent trading, typically within 90 days, by additional redemption fees [20].

Nonetheless, some investors want to consider the short-term performance of a fund, because this can help to estimate the capability of crisis management for the fund manager and his/her team, especially when an international event occurs. In addition, a number of active or impatient investors like to operate funds in a short period (e.g., several months, half a year) and favor funds with better short-term performances. Therefore, we design the short-term score function $Sscore(x, t)$ to meet the requirement, and its formula is the same with $Lscore(x, t)$ except the time period $T_S \in \{1W, 2W, 1M, 2M, 3M, 4M, 5M, 6M\}$. Due to the presentation of short-term performance, $Sscore(x, t)$ considers only weekly and monthly achievements. Here, we also adopt a GA, similar to the previous one, to fine-tune each weight $W_{T_S}(t)$.

$$Sscore(x, t) = \sum_{T_S} \frac{W_{T_S}(t)}{rank(NR_{T_S}(x, t))}.$$

Subsequently, a user-input parameter $\alpha \in [0, 1]$ is introduced to show the preference for the short-term performance, and it combines $Lscore(x, t)$ and $Sscore(x, t)$ into a score function $score(x, t, \alpha)$ as follows:

$$score(x, t, \alpha) = (1 - \alpha) \times Lscore(x, t) + \alpha \times Sscore(x, t).$$

A larger α indicates a higher interest to the short-term profitability for an investor, whereas a smaller α means the emphasis on the long-term achievement. Besides, a shorter training period makes the GA computation for $Sscore(x, t)$ faster to convergence, compared with $Lscore(x, t)$. Consequently, we can compute $Sscore(x, t)$ dynamically to report the real-time variation within a tractable time. After deciding α , one can easily obtain $score(x, t, \alpha)$ of all considered funds, and then pick up N good funds to form the portfolio.

3.4 Buying Strategy

A buying signal calls the predefined buying strategy. When receiving a buying signal, the idea in [3] is to weed out inferior funds and buy superior ones instead according to funds' ranks. In other words, they want to always keep top N funds as their investment portfolio in bull markets. Therefore, Tsai et al. used a replacement threshold $\gamma \in [0, 1]$ for weeding out more inferior funds. For example, if $|F| = 50$ and $\gamma = 0.4$, a fund $x \in F$ will be replaced by a better one once its rank falls outside top 20 ($= 50 \times 0.4$).

The problem is the difficulty in deciding γ and N simultaneously. For this reason, Tsai et al. conducted several simulations, and concluded that 10-year cumulative ROIs were located between 150% and 400% if $0.2 \leq \gamma \leq 0.9$ for various N [3]. With the conclusion, it's still hard to select appropriate γ and N , not to mention that the product of $|F|$ and γ implicitly determines N in Tsai et al. work.

To improve the shortcoming, we design a mechanism by a user-defined *tolerance coefficient* $tc \geq 1$. While a buying signal appears, we first check whether the ranks of holding funds fall outside $tc \times N$, and if yes, redeem inferior funds followed by buying superior ones instead. Figure 3 is the flowchart of our buying strategy. Furthermore, an investing behavior is expected to be conservative for setting a lower tc , because it means that the investor tends to satisfy the current ROI. On the contrary, a higher tc is selected for reducing the number of transactions.

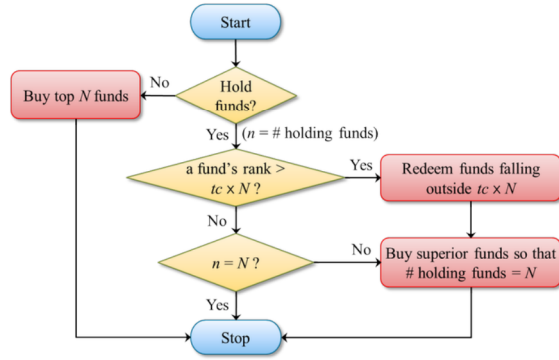


Figure 3: Buying strategy flowchart (tc is the tolerance coefficient).

3.5 Selling Strategy

From Figure 2, the investment strategy of [3] in a bear market is simply to redeem all holding funds, which somehow overreacts to the market condition. Indeed, the strategy is good for stopping loss in bear markets with internationally financial crises, e.g., 2001 U.S. 911 event and 2008 financial storm; however, it may not be practicable in a regular financial market due to a fatal drawback. That is, a volatile market may be bull and bear interleavingly so that one could purchase and redeem N funds again and again, leading to excessive transaction fees. As a result, we design a mechanism to moderate the selling procedure instead of redeeming all funds at a time. At first, an investor is requested to give two parameters,

stop-loss point $SLP \in [0, 1]$ and selling percentage $sp \in [0, 1]$. Next when a selling signal appears, we redeem $sp \times n$ inferior funds, where n is the number of holding funds at present, once the GTI has dropped under the user-specified SLP relatively to the recent climax of GTI. For convenience, we denote the ratio of $GTI(t)$ to $climax(t)$ as the *decline ratio* $DR(t)$, where $climax(t)$ is the peak value of GTI closest to day t .

As illustrated by Figure 4, while the first selling signal appears at day t_1 , we look back on previous days to find the peak GTI, i.e., $climax(t_1)$. Afterward, we find an integer $p \geq 0$ such that $SLP^{p+1} \leq DR(t_1) < SLP^p$, which is 0 in this case. Since the $DR(t_1)$ is larger than SLP^{0+1} , we do not call the selling method. As the second selling signal is triggered at t_2 , p equals to 1 and this is the first time for a selling signal appearing in the range $[SLP^{p+1}, SLP^p)$. Consequently, we redeem $sp \times n$ inferior funds according to their ranks. Besides, the strategy redeems no fund at t_3 because it is not the first selling signal in $[SLP^1, SLP^2)$. In other words, t_1, t_2, t_3, t_4 in Figure 4 are times to trigger signals, but only t_2 and t_4 call the selling strategy to redeem inferior funds. Figure 5 presents the flowchart of the selling mechanism. Note that it is similar to the selling strategy of [3] if $SLP = sp = 1$.

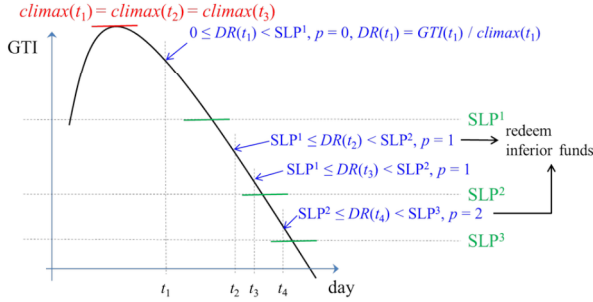
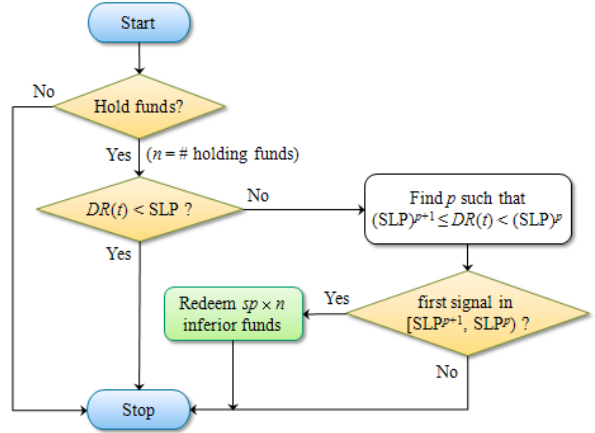


Figure 4: Times to call the selling strategy.

4 Experimental Study

In this section, we first conduct several experiments for studying the parametric settings used in our strategy, and next compare with other well-known strategies. We select equity funds with A-type and U.S. dollar, where the test data comes from FundDJ (<http://www.funddj.com>). The training and testing durations in our GA are three and five years, respectively. Since the pa-

Figure 5: Selling strategy flowchart (sp is the selling percentage).

rameters in $score(x, t, \alpha)$ refer to the data at most five years ago, 57 equity funds with available data from 1999 are selected. In other words, the testing period in our simulations is from 2007 to 2011, where the U.S. sub-prime broke out in 2008 and the European sovereign debt crisis began in late 2009. Moreover, this work adopts GAs to fine-tune all weights in $score(x, t, \alpha)$ with the population size 10. Each individual simulation is executed five times, and computes the average if only one value is shown.

We adopt a simple mechanism for the initial capital rather than specifying its value at the beginning. Since the mutual fund is a small investment vehicles, one can invest in funds with less capital relative to other investment vehicle like stocks, and we can also purchase a fund less than one unit. Therefore, our mechanism is to divide the capital evenly among desired funds in each buying procedure, where the difference between initial and current capitals is computed as the performance evaluation. Table 1 summarizes abbreviations and full names of user-specified parameters. In following simulations, we show the cumulative ROI of every three months against various para-

N	maximum number of holding funds
α	preference to the short-term achievement of funds ($\in [0, 1]$)
tc	tolerance coefficient (≥ 1)
SLP	stop-loss point ($\in [0, 1]$)
sp	selling percentage ($\in [0, 1]$)

Table 1: Summary of user-specified parameters.

metric settings, so that one can examine the profitability of our mechanism in detail. Moreover, the evaluation can help investors to take profit and start a new round of investment. Note that the cumulative ROI is to compute the returns by assuming that all holding funds are sold.

4.1 Number of Holding Funds

The parameter N is the upper bound for the number of holding funds, which could be determined by investor capital, market outlook, asset allocation, etc. Tsai et al. considered various N from 5 to 60 for studying the relationship between the cumulative ROI and the parameter for weeding out poor funds [3]. Since ICI reported that 4 mutual funds is the median number an investor owned at U.S. [1], we consider N ranging from 3 to 6 for examining their performances in Figure 6. Other configurations are the same in these simulations with $\alpha = 0$, $tc = 2$, $SLP = 0.95$ and $sp = 0.8$.

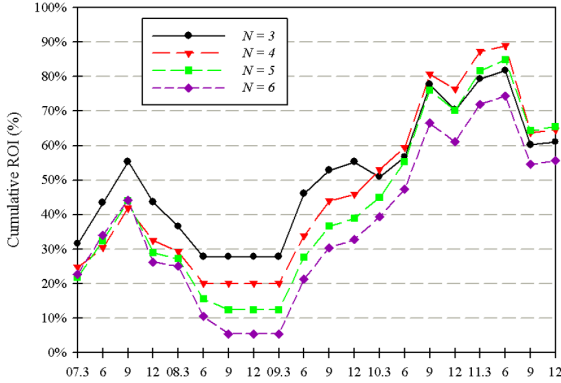


Figure 6: Cumulative ROIs with various N .

The simulation results of Figure 6 demonstrate that the performance trends of various N are almost the same. We can see that there are three significant drops in the performance: the first comes from 2008 financial storm, the second should derive from that the credit ratings of some European countries were reduced in 2010, and the third drop may continue the European sovereign-debt crisis, plus the first decision by S&P to downgrade the AAA rating of America government in 2011/8. Since these simulations adopt the setting of $\alpha = 0$ and $sp = 0.8$, their capacities to stop loss are a little poor. Nevertheless, the highest cumulative ROIs in the simulations are all over 70%, indicating that the annual profit of 14% can be achieved. An investor should specify the lock-in-gain point with a reasonable value, and redeem all holding

funds, i.e., take profit, once the desired profit is achieved.

In general, the performance decreases as N becomes large, where $N = 3$ has a decisive leading in the first three years. During the last two years, the profitability of $N = 4$ is a little higher than others.

4.2 Short-term Preference

Apart from evaluating the long-term performance of a mutual fund, some investors prefer the short-term measurement to see if the team of fund manager can response financial crises timely. A sophisticated strategy for investing in funds not only gains a high profit in bull markets, but also reduces the loss in bear ones. To take less risk, a transaction strategy should stop loss promptly as it becomes conscious of coming of bear markets. Consequently, we add the weekly and monthly considerations into our score function, and conduct corresponding simulations to study the influence of parameter α for the short-term preference.

A larger α implies the preference to short-term performance, whereas a smaller one is the emphasis on long-term achievement. We also fix other configurations in these simulations with $N = 4$, $tc = 2$, $SLP = 0.95$ and $sp = 0.8$ to evaluate the effects of different α .

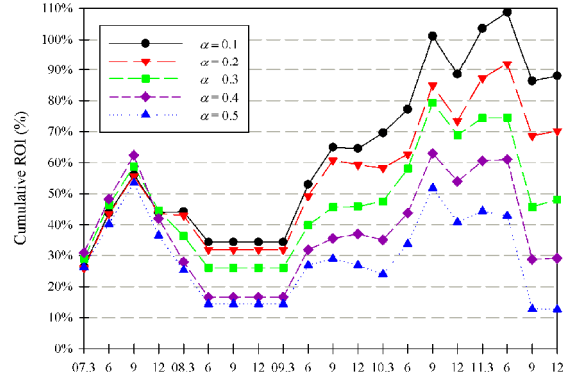


Figure 7: Cumulative ROIs with different α .

From the experimental results in Figure 7, one can see that the α setting could make quick adaption to the changed financial situations, regardless of achieving the high profit or reducing the risk. Compared to the simulations in Figure 6, the configurations of $\alpha = 0.1$ and 0.2 have better performances to stop loss in 2008. On the contrary, simulations with larger α have much poor achievements. The reason would be the rising cost on the

transaction fee for a larger α , which will become clear later.

4.3 Investor Attribute

Understanding the investment attribute of the investor himself is useful to receive a satisfied profit. Many financial firms provide questionnaires to evaluate investors' attributes so as to supply appropriate investment tactics and vehicles. According to the common risk levels, there are three primary styles of investment attributes: *conservative*, *stable* and *active*. A conservative investor expects less risks and can't endure a sharp fluctuation on ROI, an active one can accept higher risks to make potential profits, whereas a stable one is something in between. Under different financial situations, investors with different attributes have distinct investing actions, and our mechanism can reflect investors' attributes by using parametric settings. Figure 8 is the simulation results considering different sp for three types of investment attributes: $sp = 0.9, 0.7$ and 0.5 for representing conservative, stable and active investors, respectively. Similarly, other parameters of $N = 4$, $\alpha = 0.1$, $tc = 2$ and $SLP = 0.95$ are all fixed.

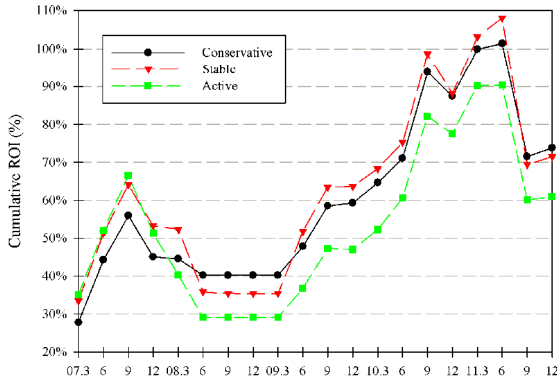


Figure 8: Cumulative ROIs with different investor's attributes.

We expect that a configuration for simulating the conservative behavior could maintain the profit in bear markets, and in the three significant drops in Figure 8, it indeed has a better performance than other settings. The active setting of $sp = 0.5$ in Figure 8 would not redeem funds for a straw in the wind of the financial market, so that it can save more transaction fees than the conservative setting. Interestingly, the setting for simulated the stable investor gains a better profit in

the capricious market from 2009, where European sovereign debt crisis began.

4.4 Amount of Transactions

The cost of transactions has a great effect to obtain a satisfied profit. An excessive transaction cost could eat into the return to show even a negative remuneration, where the transaction cost is composed of two parts: the fee of each transaction and the amount of transactions. Each transaction fee consists of several charges, including management, front-end and back-end loads, etc. It is difficult to set detailed charges for a transaction, since there is no standard for these fees. Moreover, each charge of a transaction could be different among mutual funds, selling agents, or even transaction days (e.g., in the preferential period). As a result, we fix the transaction fee to 3% in all simulations, which is not only close to the total cost given by [5] but also convenient for the comparison to [3].

Figure 9 shows the amount of transactions in the simulations of Figure 7. It is not surprised that a larger α generates more transactions, and hence, causes a worse performance than a smaller α has. Note that we don't conclude that a small transactions is a required condition for making a high return. In fact, parts of our experiments achieve good performances even though they have a little high amounts of transactions. However, it is definitely that a small number of transactions assists an investor in reducing the transaction cost.

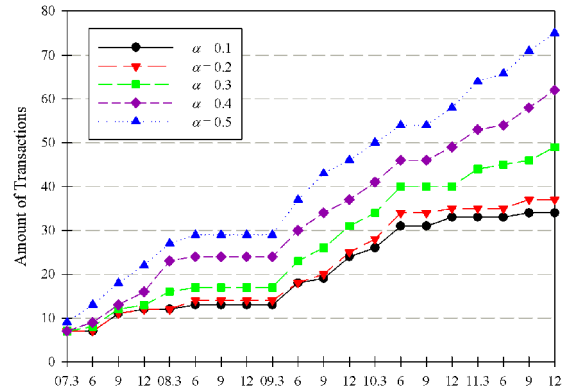


Figure 9: Amount of transactions to different α .

4.5 Comparison with Other Methods

This section compares our investment scheme with other strategies by considering $N = 4$. We introduce the *4433 rule* (FundDJ), a simple and

well-known investment strategy in Taiwan, as one of compared strategies. Its main idea is to pick the funds with both better short-term and long-term performances of the same types in the market. For all funds with the same type, 4433 rule is as follows: the first “4” picks the funds with one-year achievement of top quarter; the next “4” is to keep the funds with two-, three- and five-year achievements of top quarter; “33” is to filter out those funds whose six- and three-month achievements are not inside top third. In general, there are very few funds which can meet 4433 rule in a type, and the number of funds we bought is less than N for no desired funds. Provided that there are too many funds satisfying 4433 rule, we further compare their one-month performances to select N funds at most. The experiment of 4433 rule is to evaluate the portfolio monthly, and weed out funds against the rule followed by buying conformed ones.

In addition, our mechanism is slightly modified for simulating two investment methods. At first, for simulating 4433 rule, we fix all weights in $score(x, t, \alpha)$ to be 1 if their time periods are used by 4433 rule, and 0 otherwise. Other parameters are set to be identical as the stable investment setting in Figure 8, and the method is named *simulated 4433*. Next, we attempt to simulate the work [3], and replace our GTI definition by theirs.

The parametric settings are $\alpha = 0$ and $tc = SLP = sp = 1$, where the derived method is called *simulated Tsai's work*. Furthermore, we use the result of stable investment in Figure 8 for comparing with aforementioned strategies.

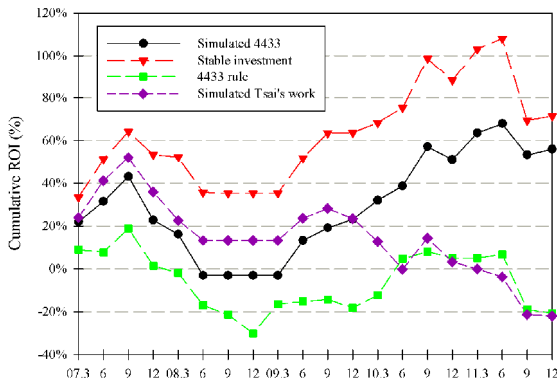


Figure 10: Comparison of different investment strategies.

Figures 10 and 11 are the experimental results of different investment strategies and their amount of transactions, respectively. In Figure 10, the

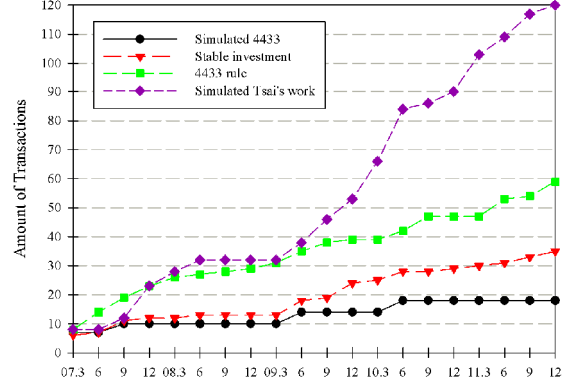


Figure 11: Amount of transactions for strategies in Figure 10.

profitability of 4433 rule is not as good as we expect. The reason may result from the capriciously financial situation in the testing duration. Figure 11 also demonstrates a little high number of transactions for 4433 rule, indicating that the holding funds are changed frequently. Nevertheless, the number of transactions of our simulated 4433 is less than half of 4433 rule, and it also makes much better performance. In addition, simulated Tsai's work has good profitability before 2009, and thereafter, its amount of transactions is rocketing upwards, which should be the main reason to get poor performance. Finally, its cumulative ROI in the end of testing duration is scarcely different to that of 4433 rule.

From Figure 11, though the setting of stable investment has a little higher amount of transactions than that of simulated 4433, it makes much better returns in Figure 10. Despite bull or bear markets, the profitability of stable strategy takes a runaway lead in the whole testing period. Moreover, it stops the loss at 35.4% and gains the highest ROI of 108%.

5 Conclusion

In the age where the real interest rate is negative, it's tremendously important to keep the value of assets through the financial investment. Whereas the mutual fund has been one of the popular investment vehicles, we develop strategies based on the genetic algorithm for investing in funds. Experimental results exhibit that our strategy can not only achieve the high profit, but also reduce the risk concurrently. In addition, our improvement on the buy-and-sell strategy can signif-

icantly reduce the amount of transactions, which saves transaction fees accordingly.

Moreover, the investing attribute is major indicator for evaluating one's risk tolerance. An investor can input parameters according to the investing attribute of himself, where these parameters guide the actions of our buy-and-sell strategy to fulfil the risk attitudes of investors. Extensive simulation results show that our strategy for mutual fund investment can also meet investor's attribute. For investors who want to invest in funds but are hard to select proper ones, our work could give satisfactory references for investing decisions.

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